Algebra I Statistics and Probability Cheatsheet

Module 11

Measures of Center

$$\text{mean} = \frac{\text{sum of data values}}{\text{number of data values}}$$

Median is the middle value after ordering the data. Mode is the most frequent value.

Measures of Spread

$$\text{range} = \max - \min$$

$$\text{IQR} = Q_3 - Q_1$$

Five-number summary:

$$\min, \quad Q_1, \quad \text{median}, \quad Q_3, \quad \max$$

Box Plots

- Box runs from Q_1 to Q_3.
- Median line is inside the box.
- Whiskers extend to minimum and maximum unless outliers are separated.

Outlier rule:

$$x < Q_1 - 1.5(\text{IQR}) \quad \text{or} \quad x > Q_3 + 1.5(\text{IQR})$$

Two-Way Tables

Rows and columns show categories. Joint frequencies are inside the table. Marginal frequencies are row or column totals.

$$P(A) = \frac{\text{count in } A}{\text{total count}}$$

$$P(A \mid B) = \frac{\text{count in both } A \text{ and } B}{\text{count in } B}$$

Basic Probability

$$P(\text{event}) = \frac{\text{favorable outcomes}}{\text{total outcomes}}$$

$$0 \le P(A) \le 1$$

Complement:

$$P(\text{not } A) = 1 - P(A)$$

Compound Probability

Addition rule:

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

Independent multiplication rule:

$$P(A \text{ and } B) = P(A)P(B)$$

Odds

$$\text{odds in favor} = \frac{\text{favorable outcomes}}{\text{unfavorable outcomes}}$$

$$\text{odds against} = \frac{\text{unfavorable outcomes}}{\text{favorable outcomes}}$$

Quick Checks

- Order data before finding median or quartiles.
- Use totals carefully in two-way tables.
- Probabilities can be fractions, decimals, or percents.
- "Or" includes overlap unless the events are mutually exclusive.